INTERNETWORKING ESSENTIALS CA1

BACHELOR OF TECHNOLOGY

IN

Computer Science & Engineering

By

Mullagura Harsha Vardhan Naidu

SECTION-K23UP

Roll no: 08

Reg.No: 12302595

TO

Mr. Singh Malhi Sir

LOVELY PROFESSIONAL UNIVERSITY

PUNJAB INDIA

Project8: _You are hired as a network engineer for thenetwork Infotech Ltd., a mid-sized enterprise with a 10-floor office building Each floor is equipped with 5 computers, and the organization requires a well-structured network to ensure efficient communication and scalability.

Network Design Requirements:

1. Topology Selection: Design a star topology for 5 floors, a ring topology for the 3 floors, and then a mesh topology for 2 floors, considering performance and fault tolerance
2. IP Addressing Scheme: The company has decided to use Class C private IPv4 addresses following a classful addressing scheme Allocate IP addresses properly for each floor, ensuring uniqueness
3. Routing Strategy for Inter-Floor Communication & Connectivity: Recommend: 2 routing approach that is dynamic for inter-1 floor communication.

Design how the floors will be connected for seamless inter-department communication

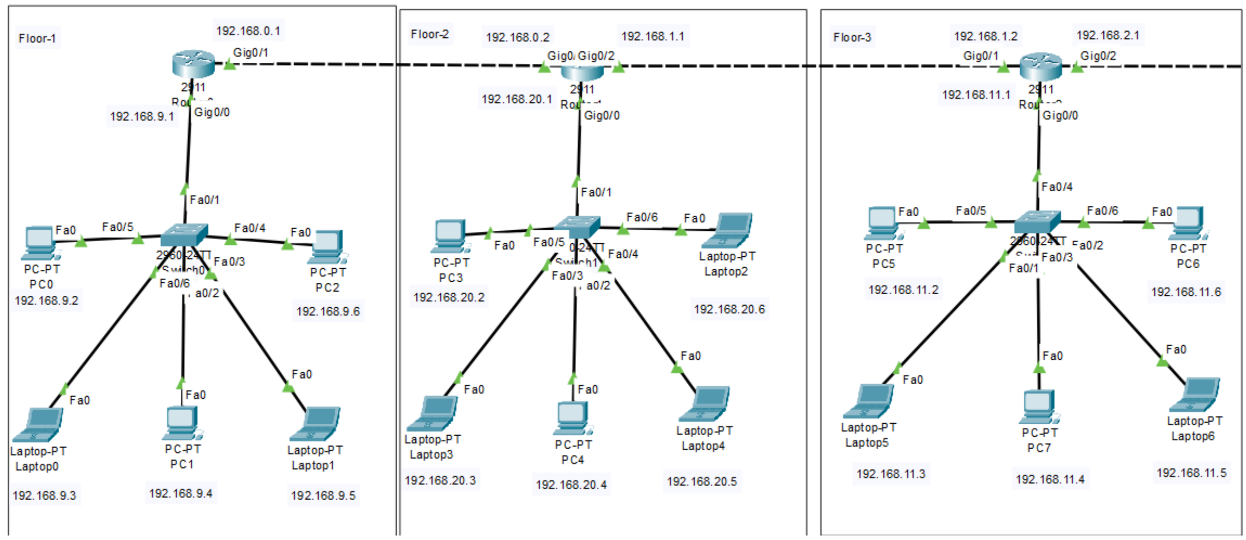
Suggest the appropriate network devices (e..., switches. routers, access poits) and their placement.

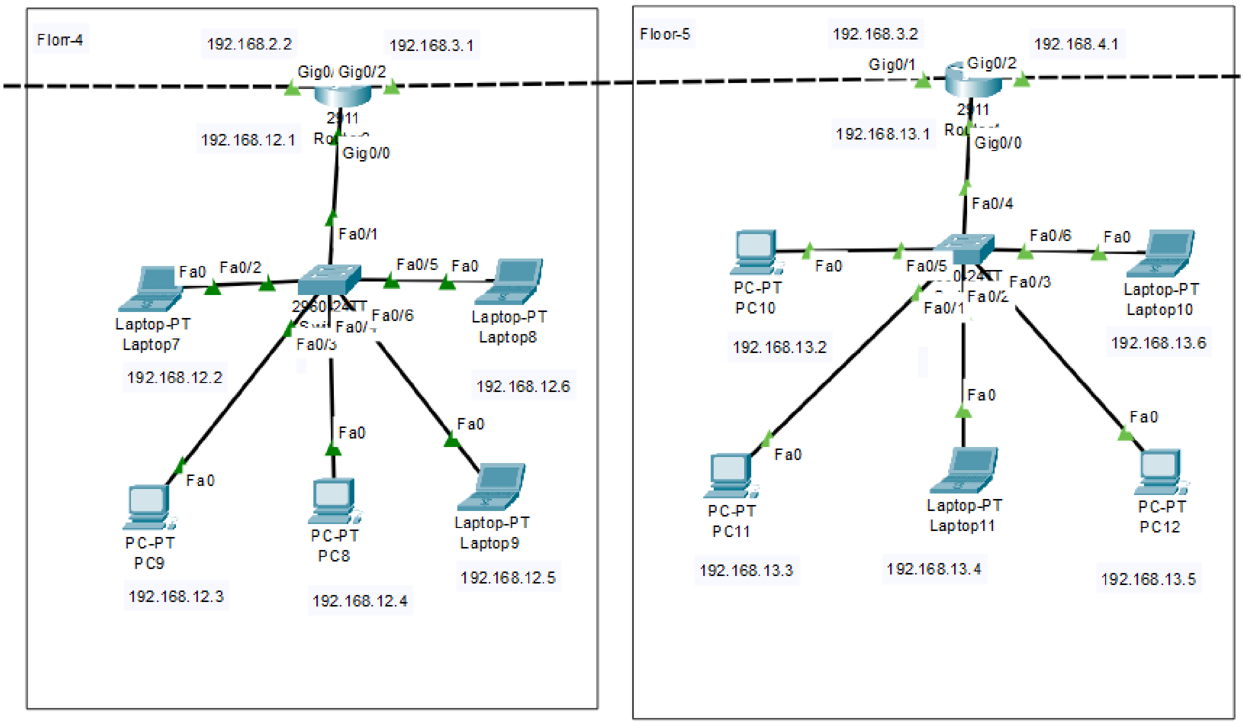
If using dynamic routing, suggest an appropriate routing protocol (e..., RIP, OSPF, or EIGRP) with justification.

If using static routing, define the static routes for efficient data flow.

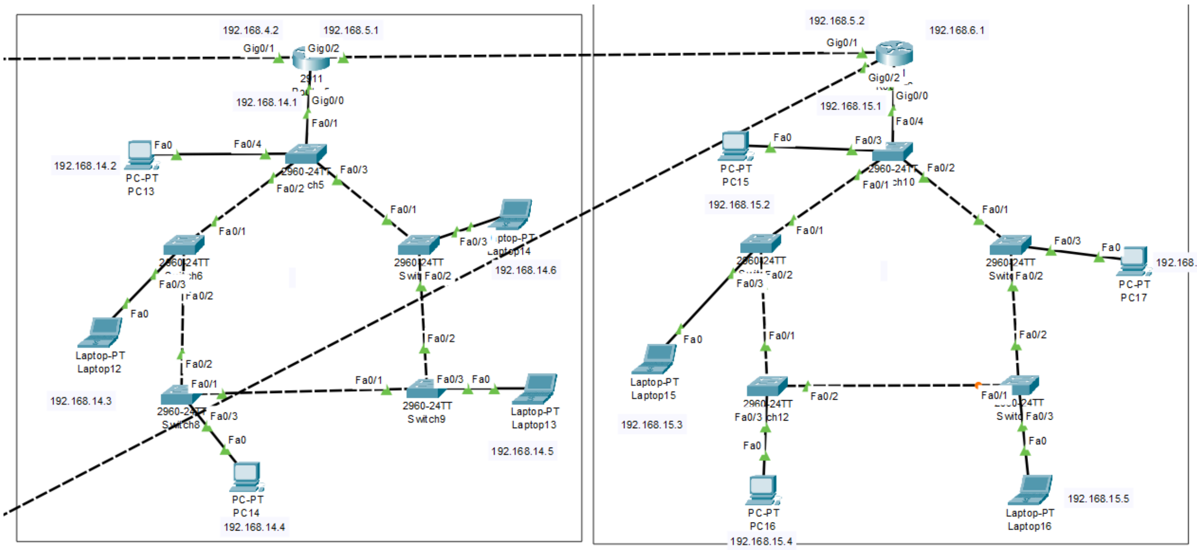
Specify the number of default gateways along with IP addresses.

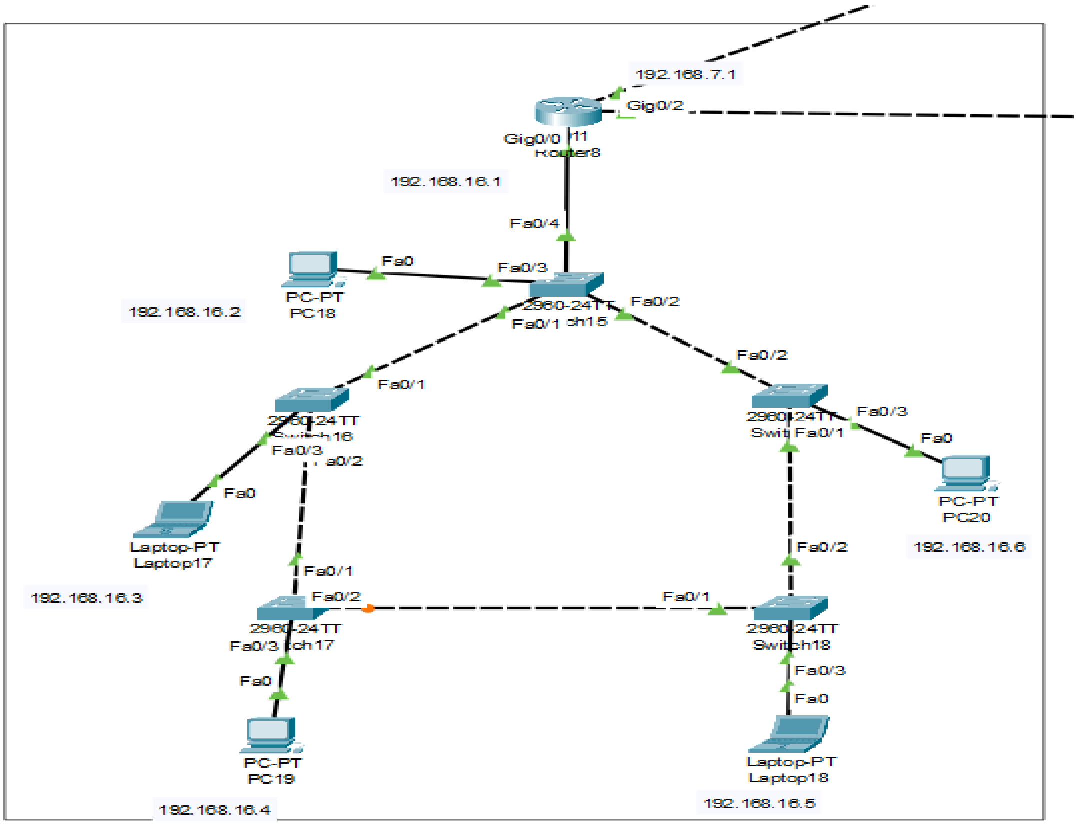
Star Topology for first 5 floors:-



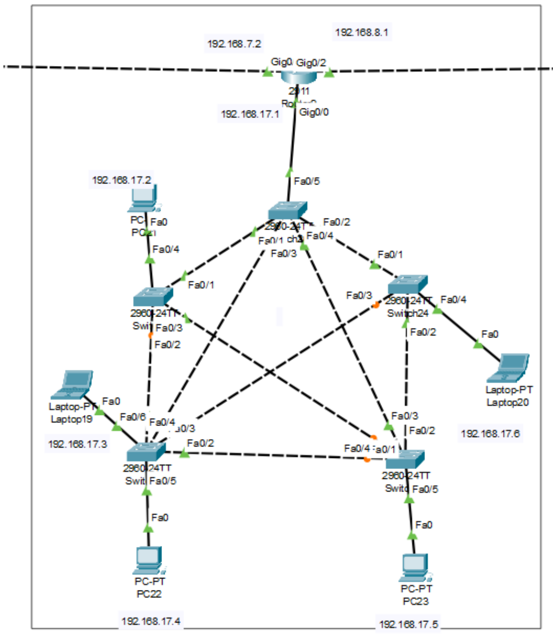


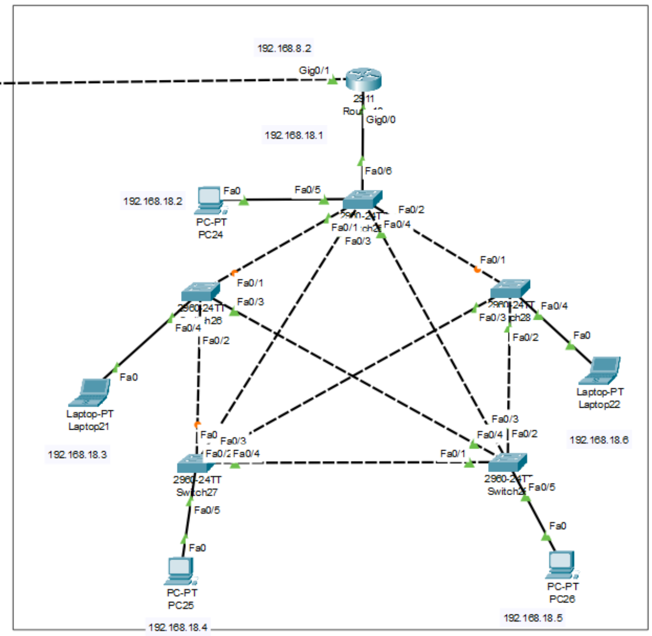
Ring Topology for next 3 floors:-



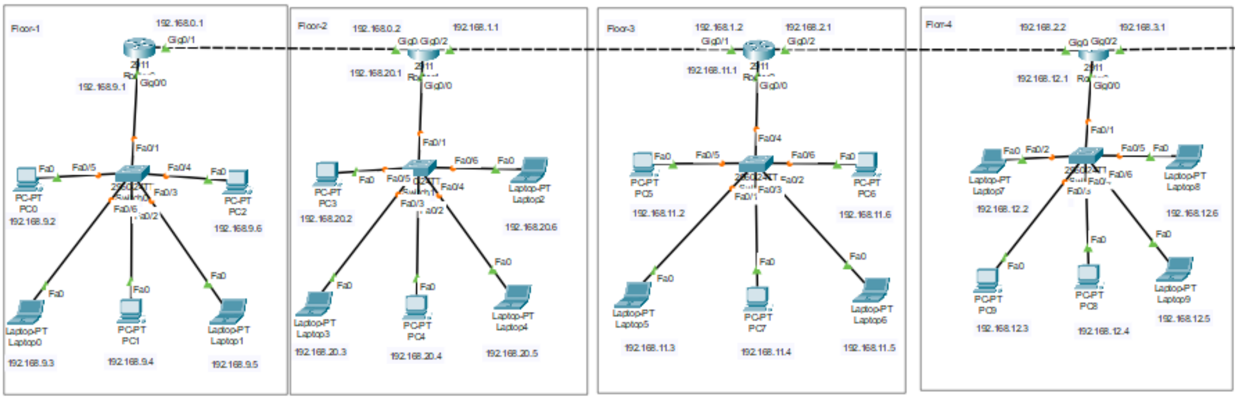


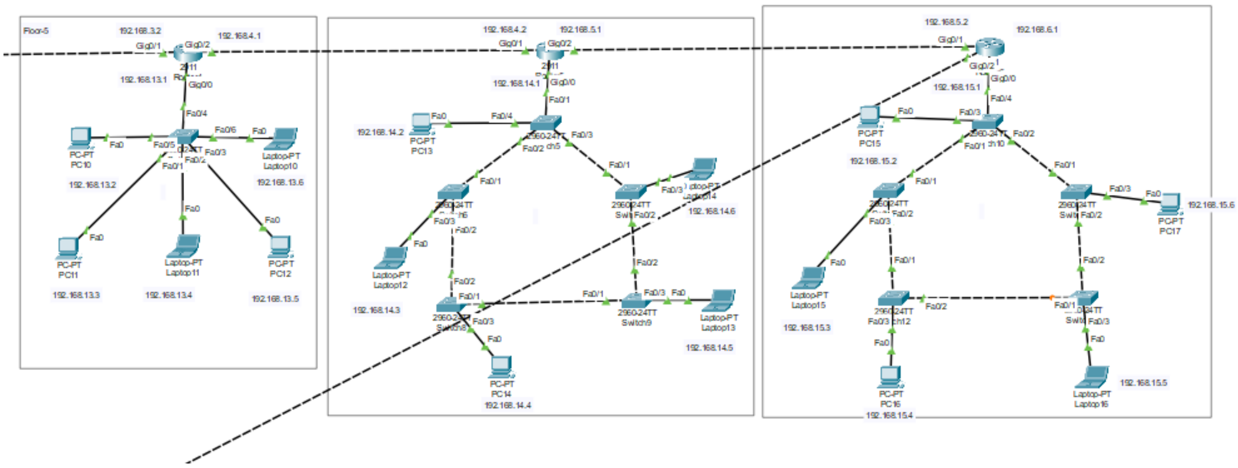
Mesh topology for Last 2 floors:-

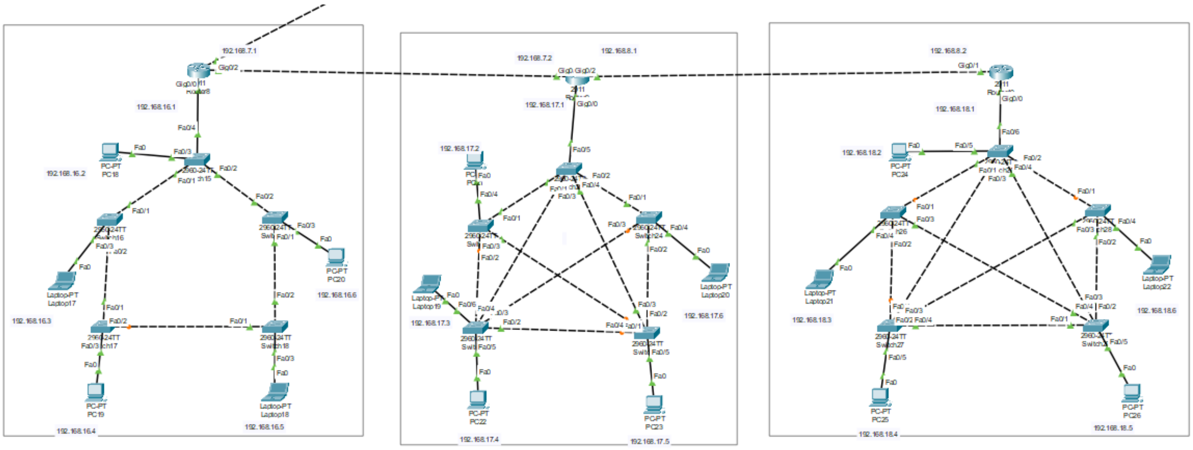




Entire building with complete setup

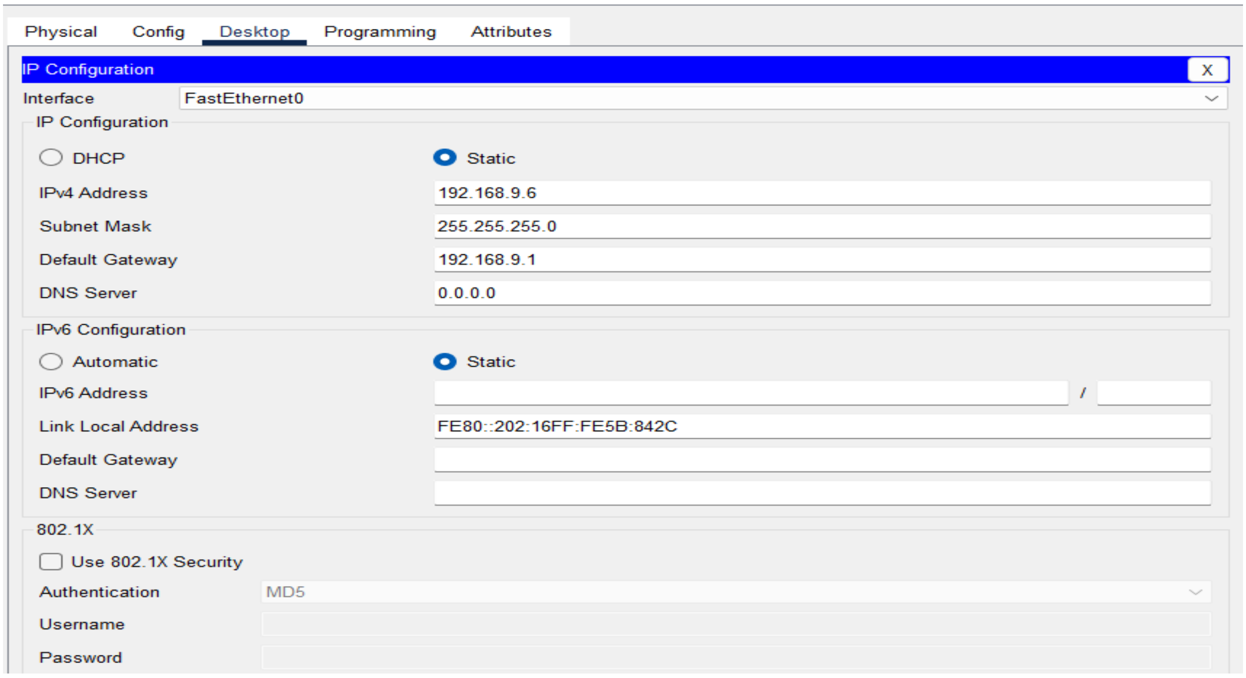




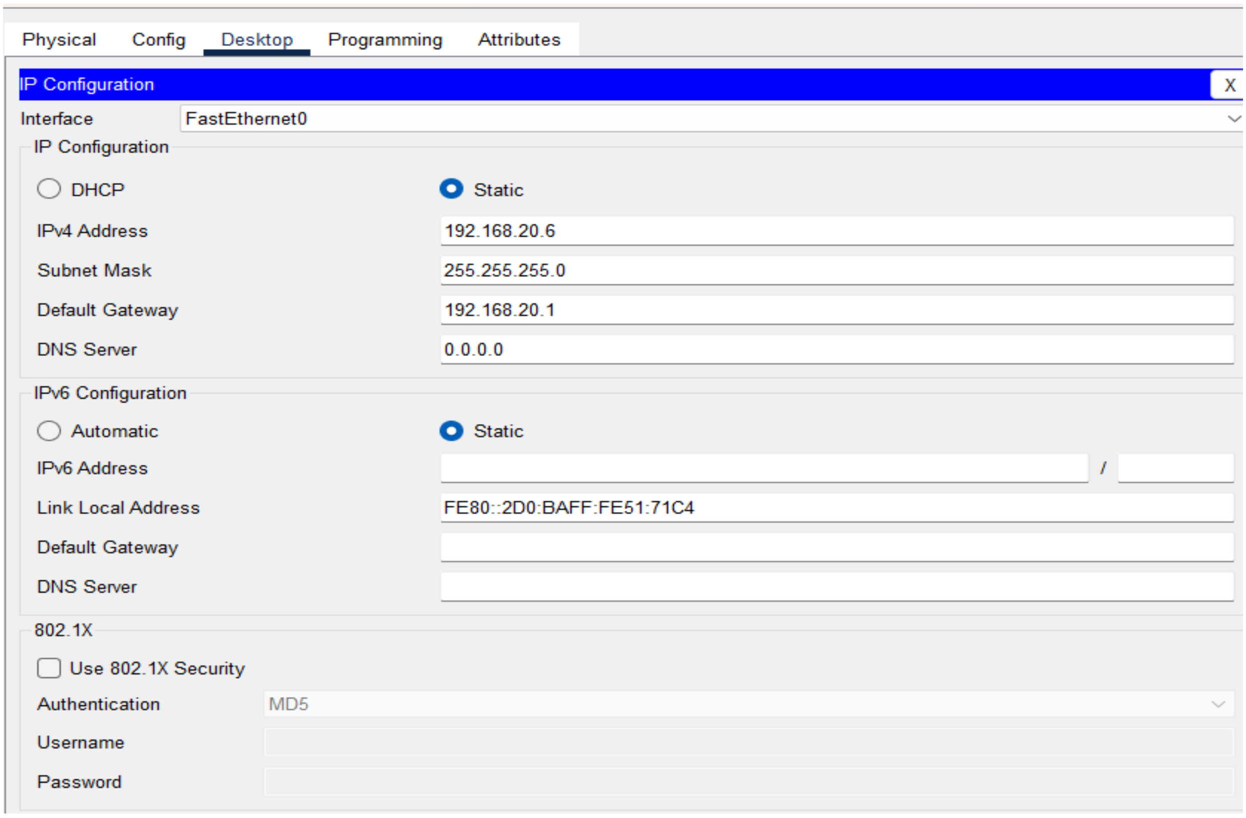

192.168.7.1
Gig0/2
192.168.16.1
Fa0/4
Fa0
Fa0/3
PC-PT
PC18
192.168.16.2
Fa0/2
Fa0/1
Laptop-PT
Laptop17
192.168.16.3
2960-24TT
Switch18
Fa0/3
PC-PT
PC20
192.168.16.6
PC-PT
PC19
192.168.16.4
Laptop-PT
Laptop18
192.168.16.5
192.168.7.2
192.168.8.1
2911
192.168.17.1
Gig0/0
Fa0/5
192.168.17.2
Switch24
Laptop-PT
Laptop19
192.168.17.3
Laptop-PT
Laptop20
192.168.17.6
PC-PT
PC22
192.168.17.4
PC-PT
PC23
192.168.17.5
192.168.8.2
Gig0/1
192.168.18.1
Fa0/6
192.168.18.2
PC-PT
PC24
Laptop-PT
Laptop21
192.168.18.3
Laptop-PT
Laptop22
192.168.18.6
Switch27
PC-PT
PC25
192.168.18.4
PC-PT
PC26
192.168.18.5

1. Allocation of IP Address:

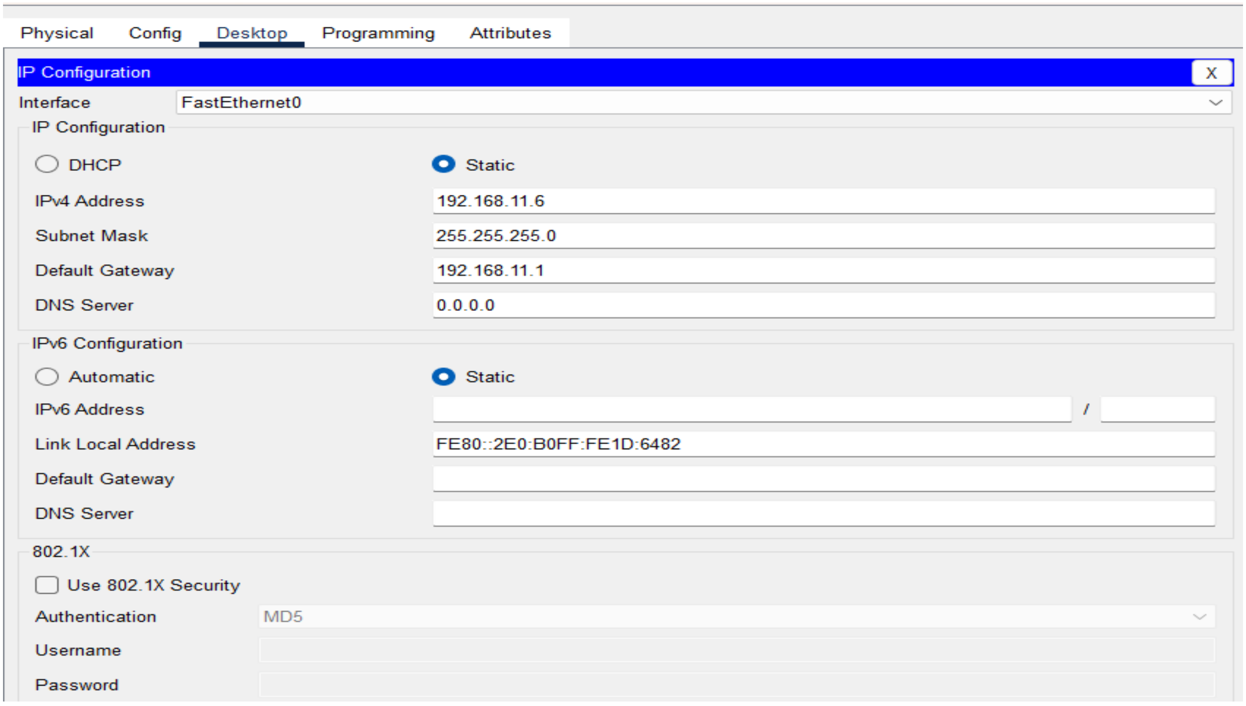
1st Floor:



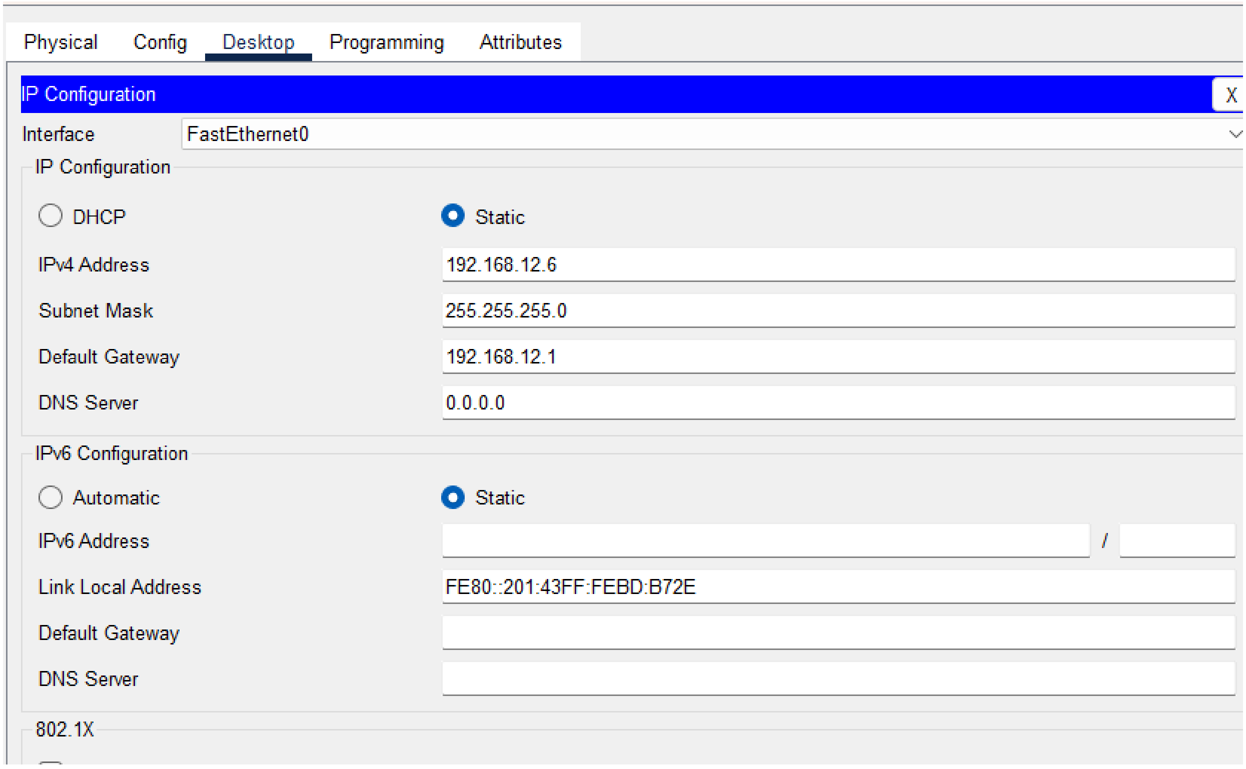
2nd Floor:



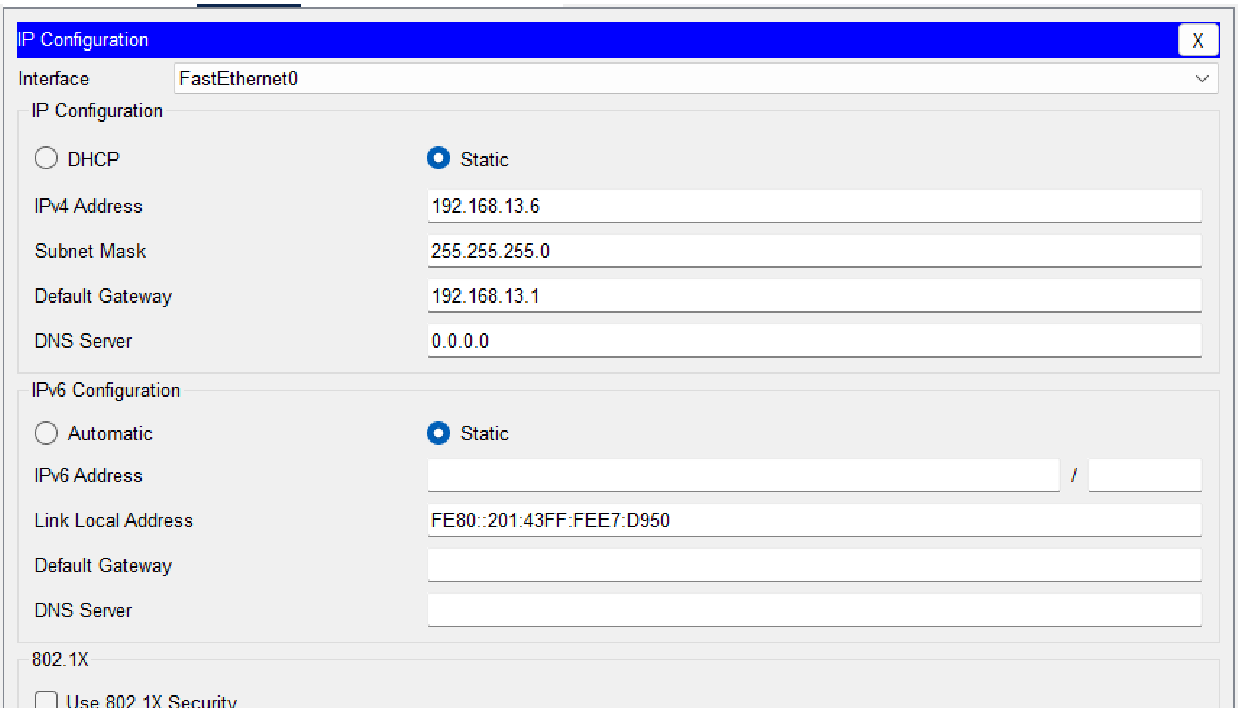
3rd Floor:



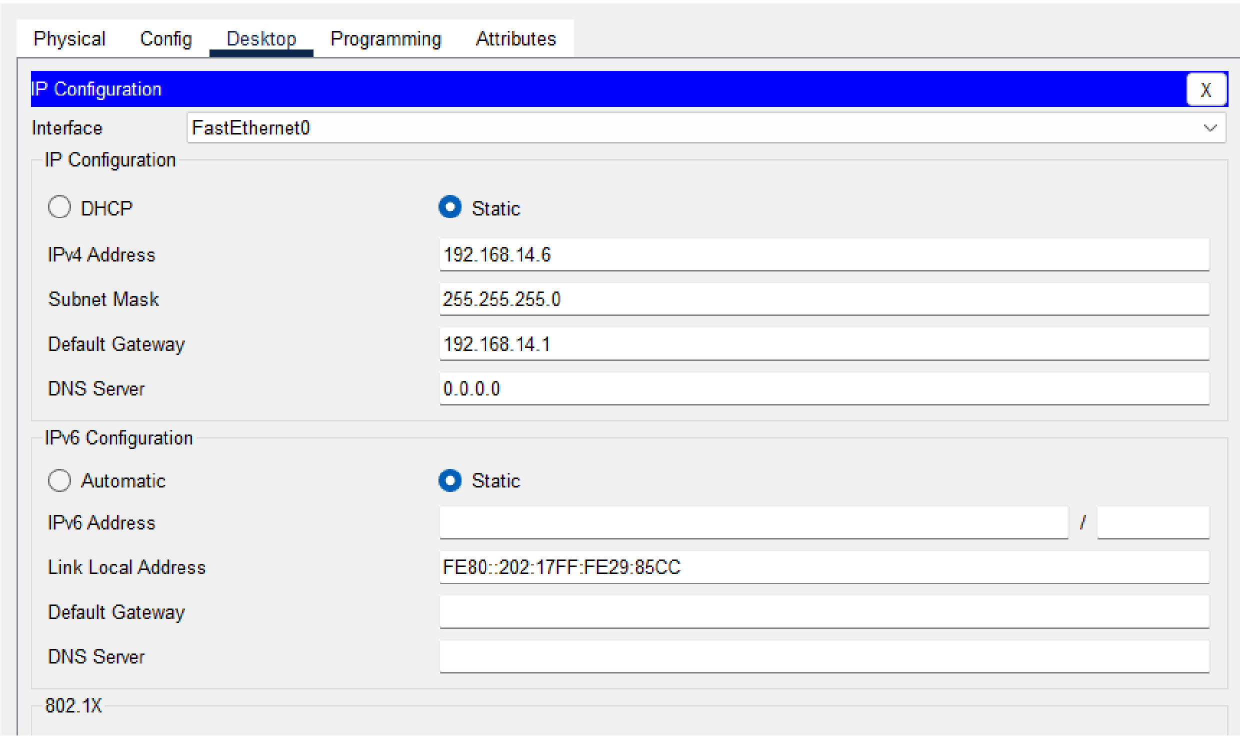
4th Floor:



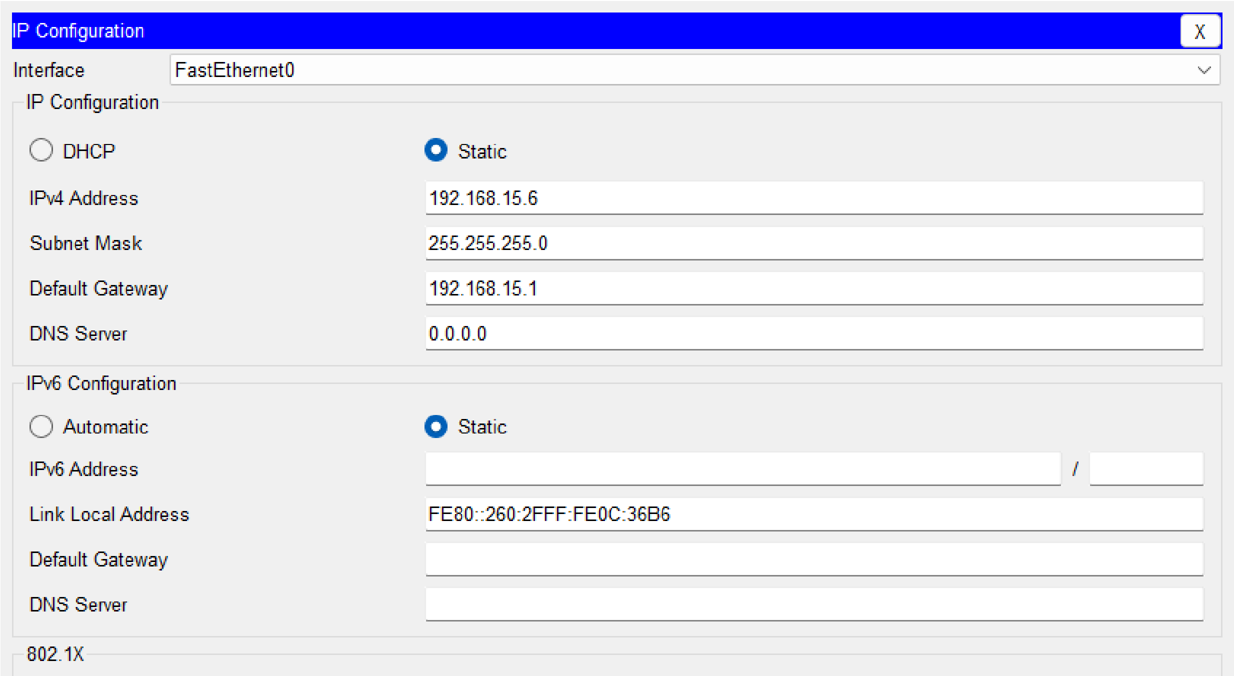
5th Floor:



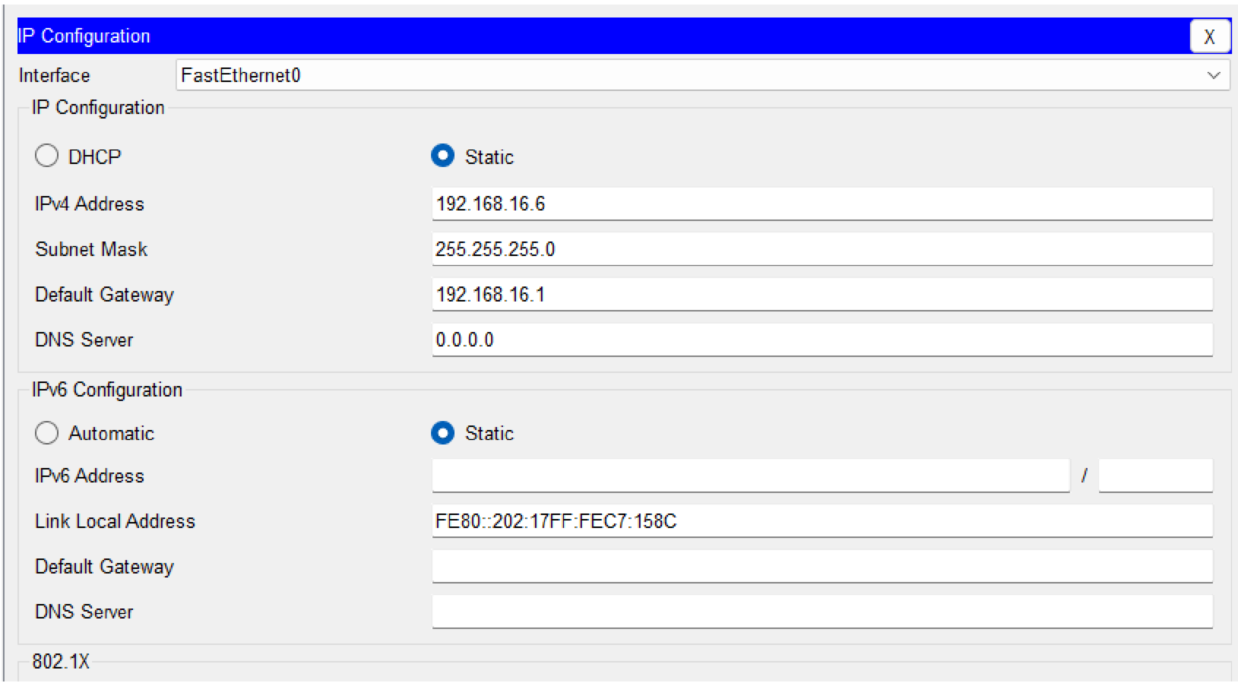
6th Floor:



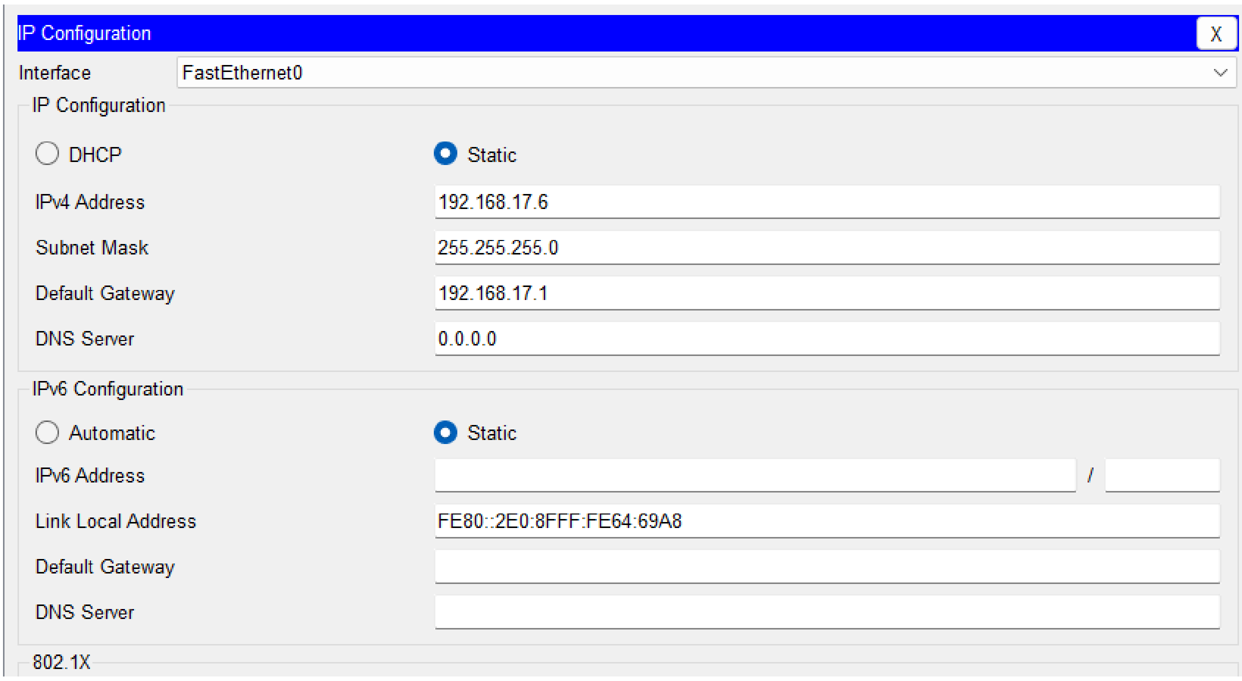
7th Floor:



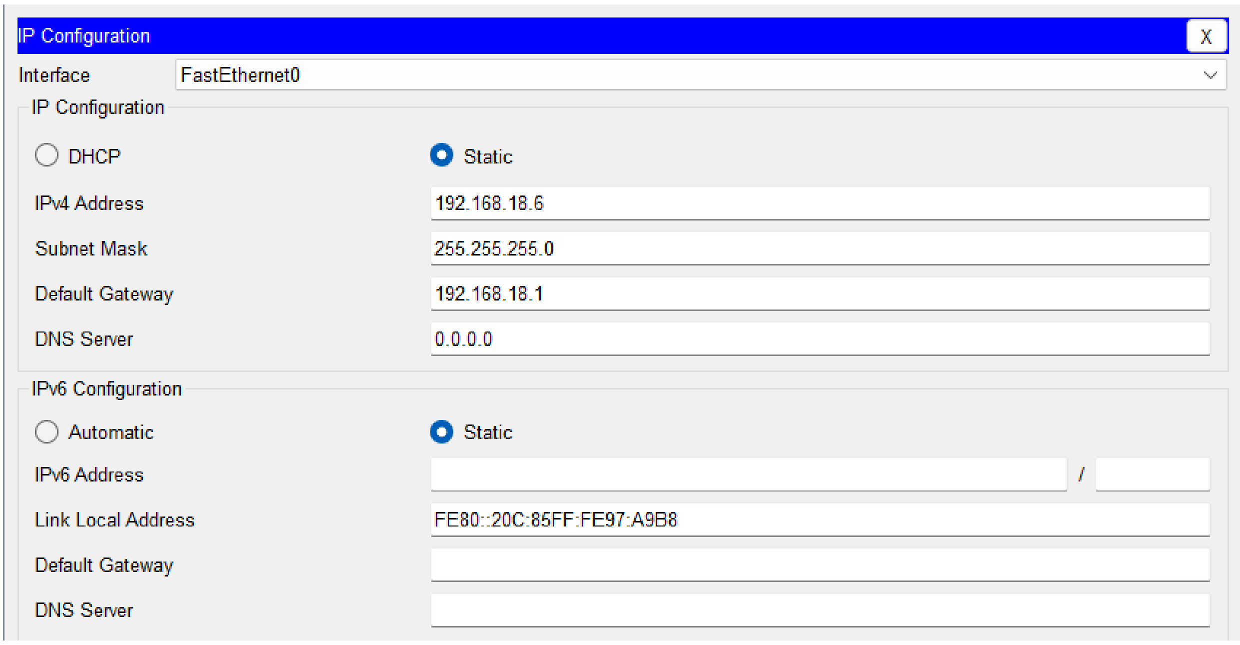
8th Floor:



9th Floor:

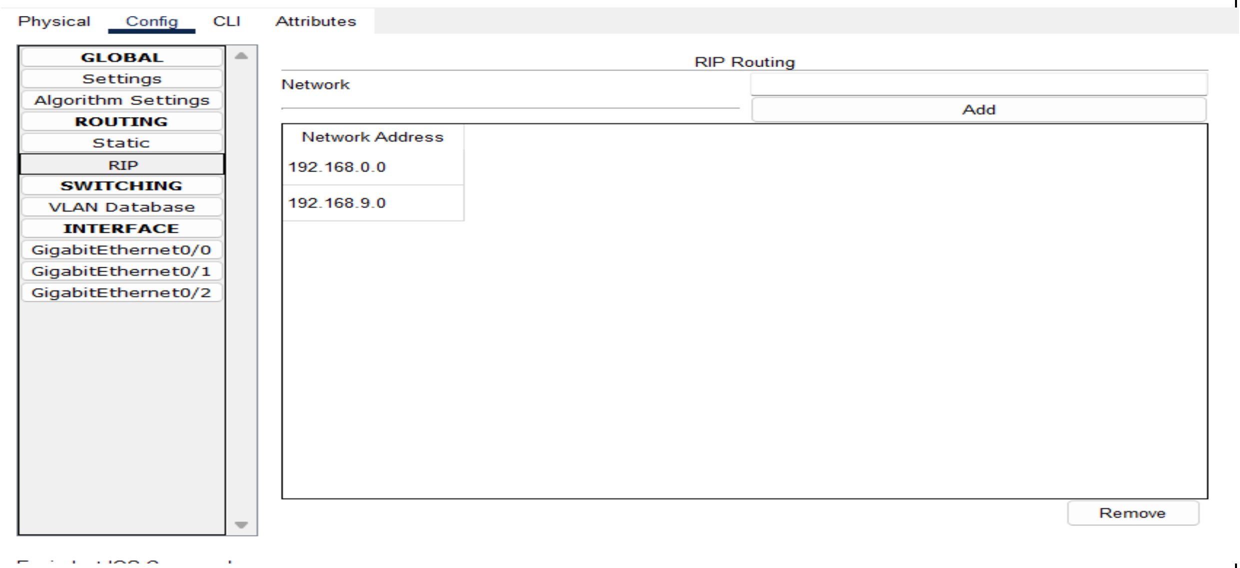


10th Floor:

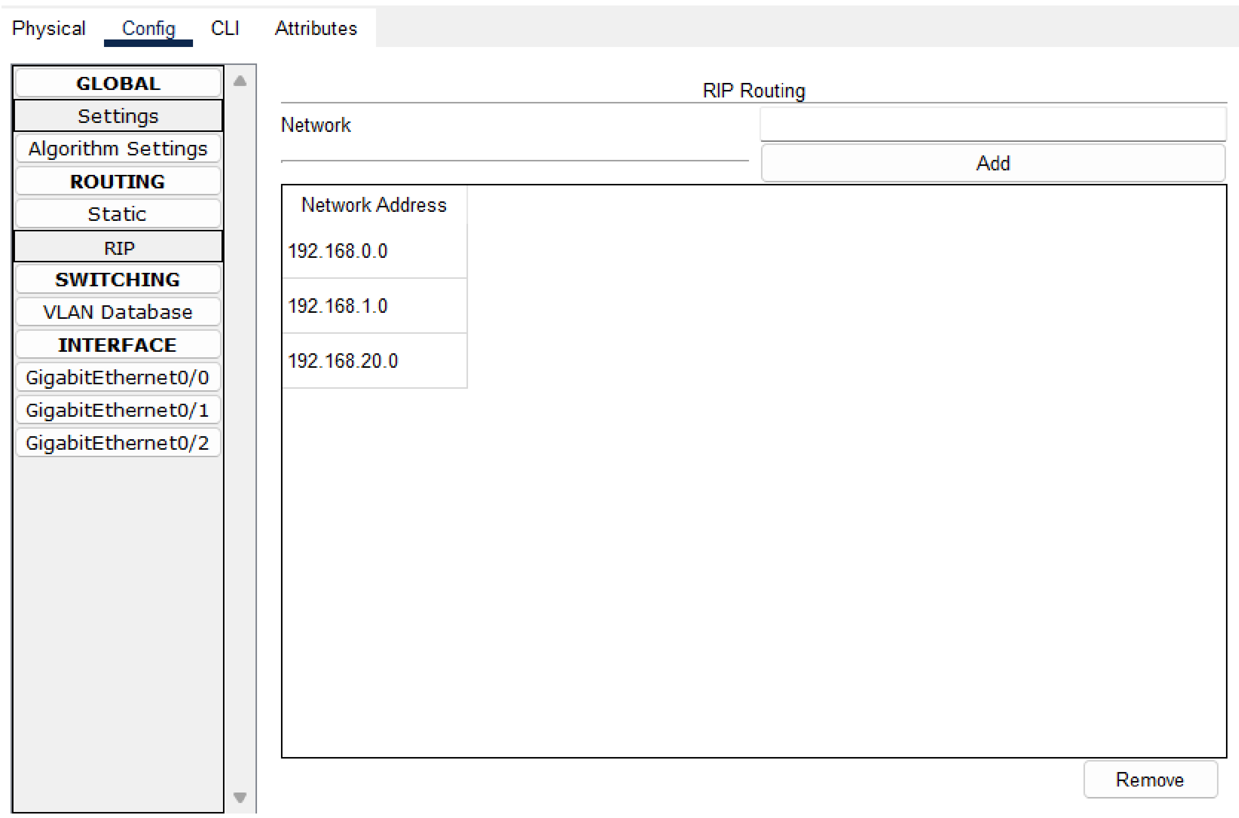


2. Dynamic Routing:

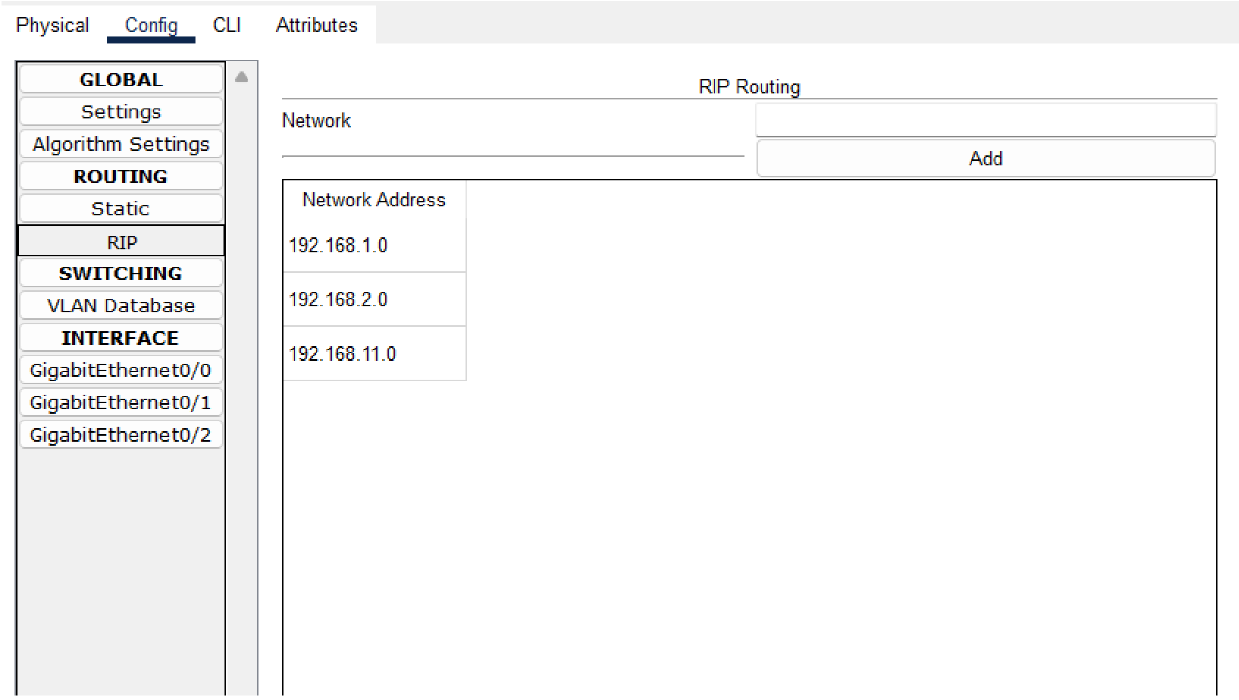
1st Router:



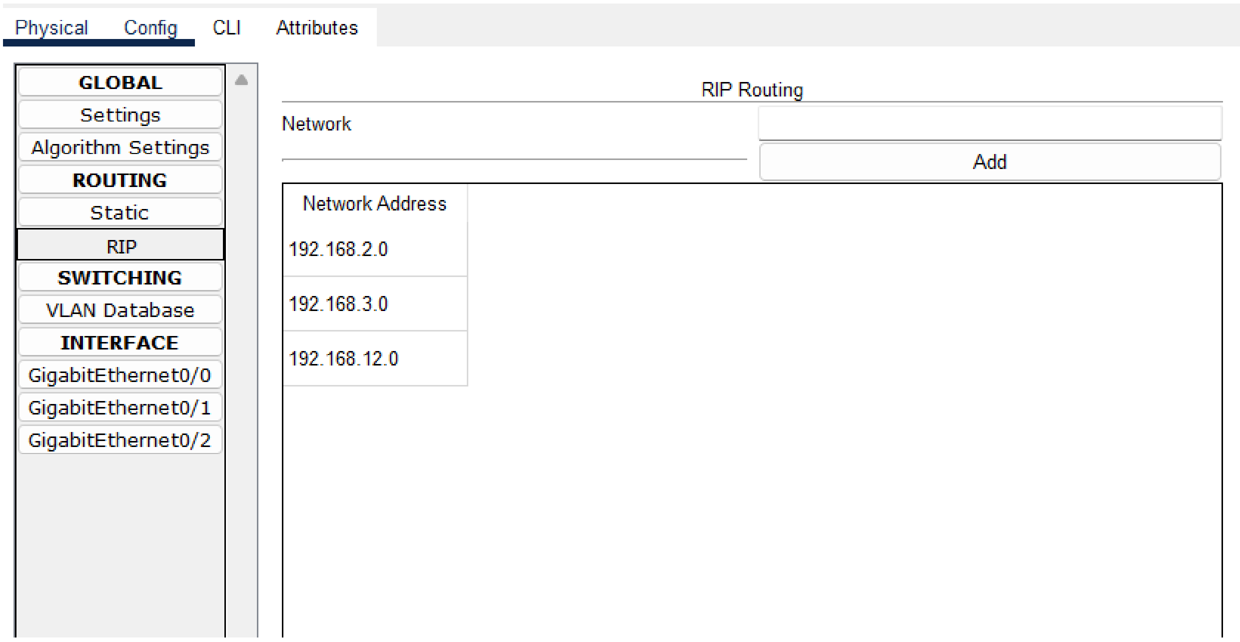
2nd Router:



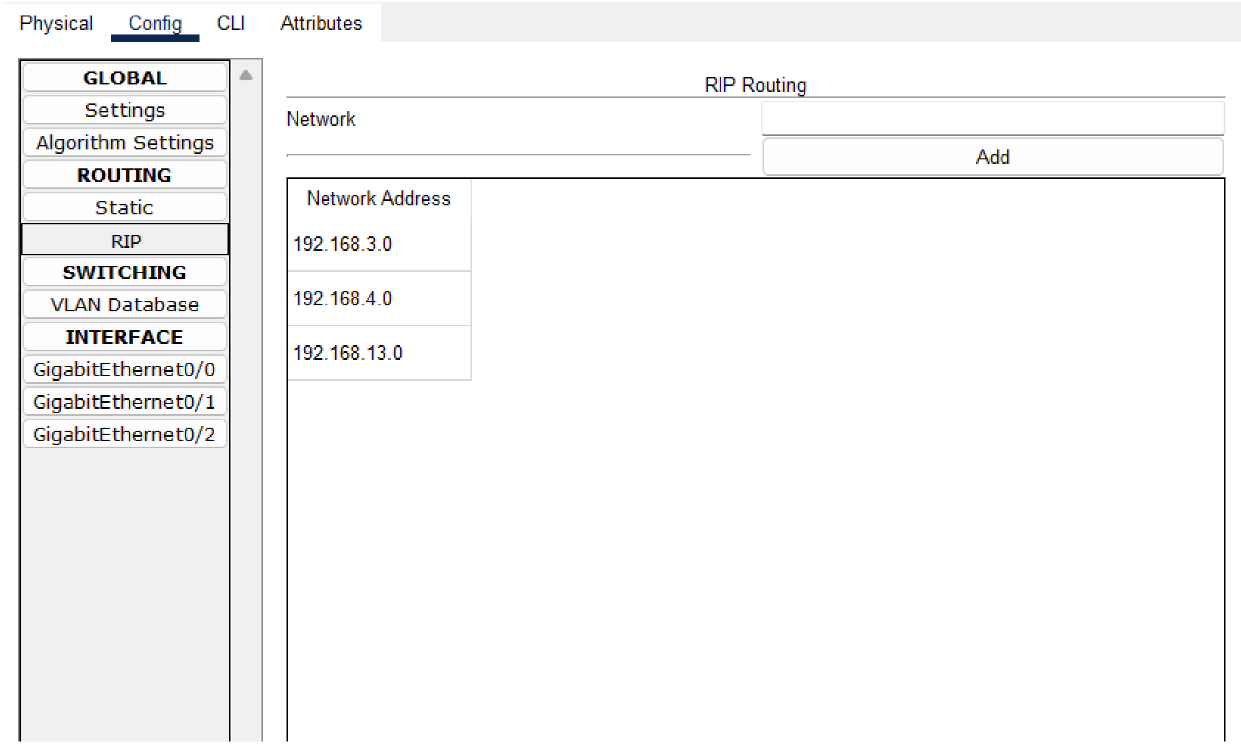
3[rd] router:



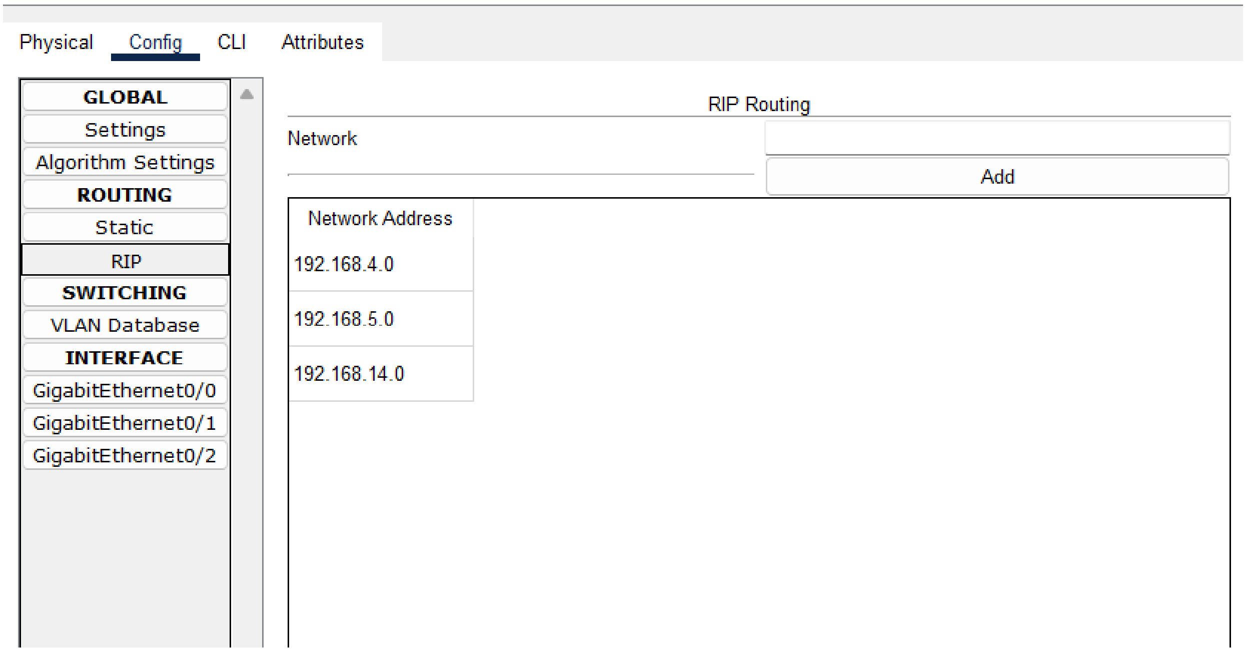
4[th] router:



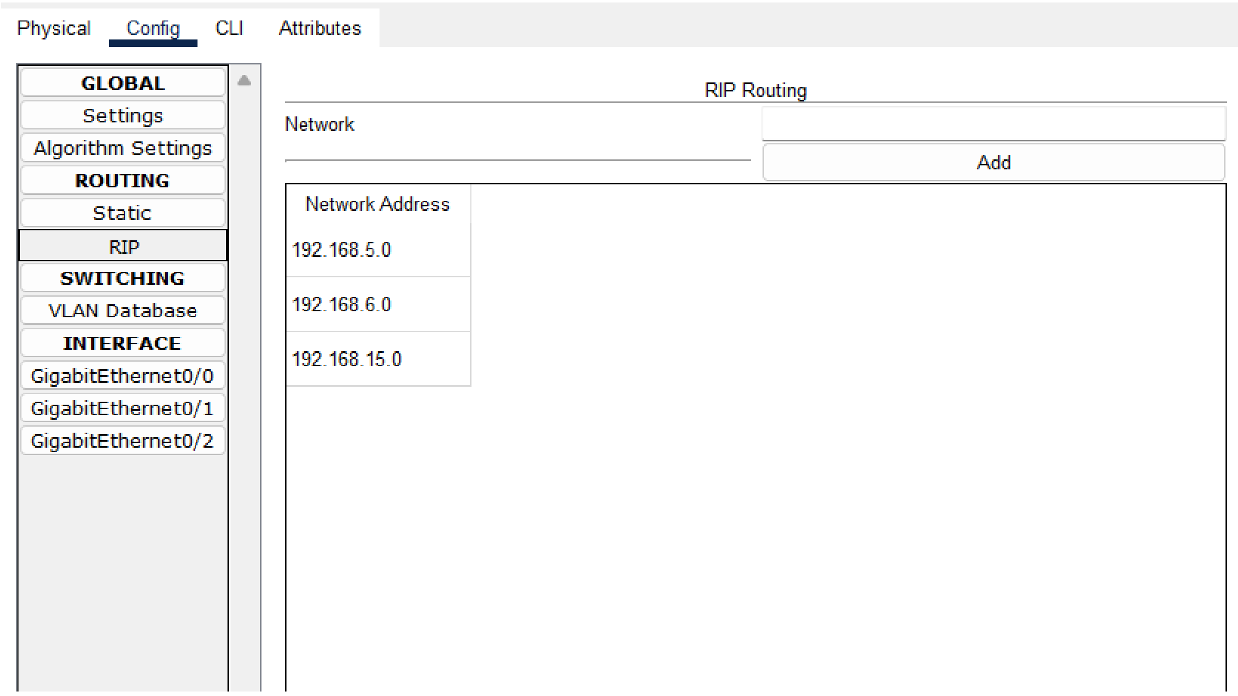
5th router:-



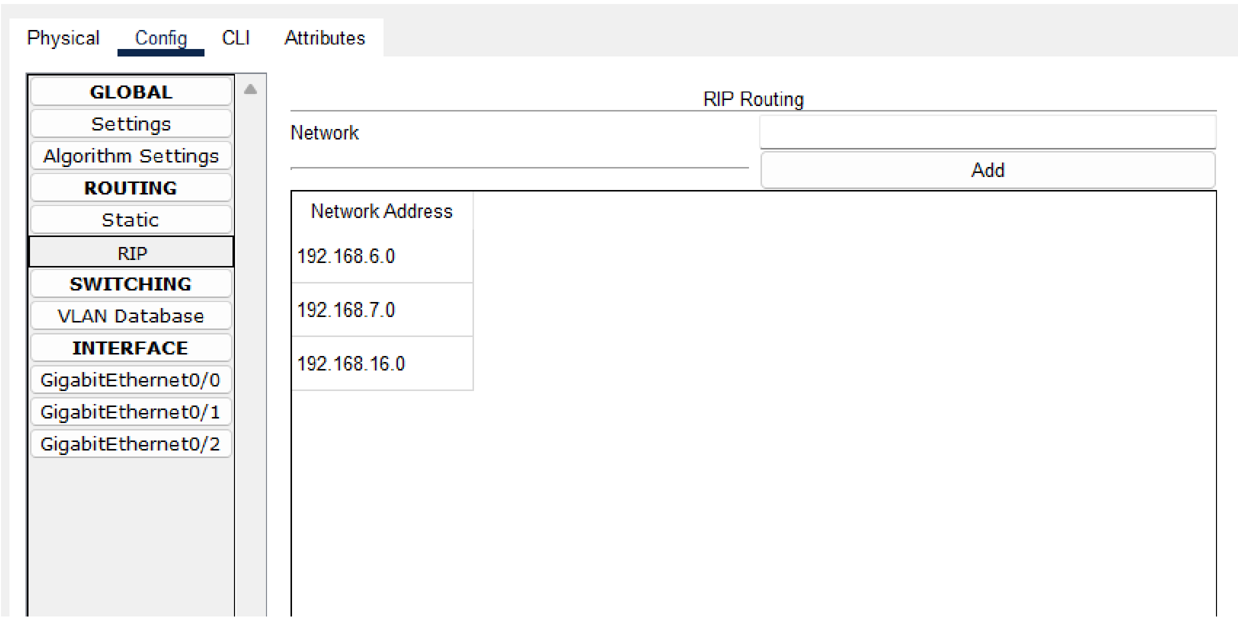
6th router:-



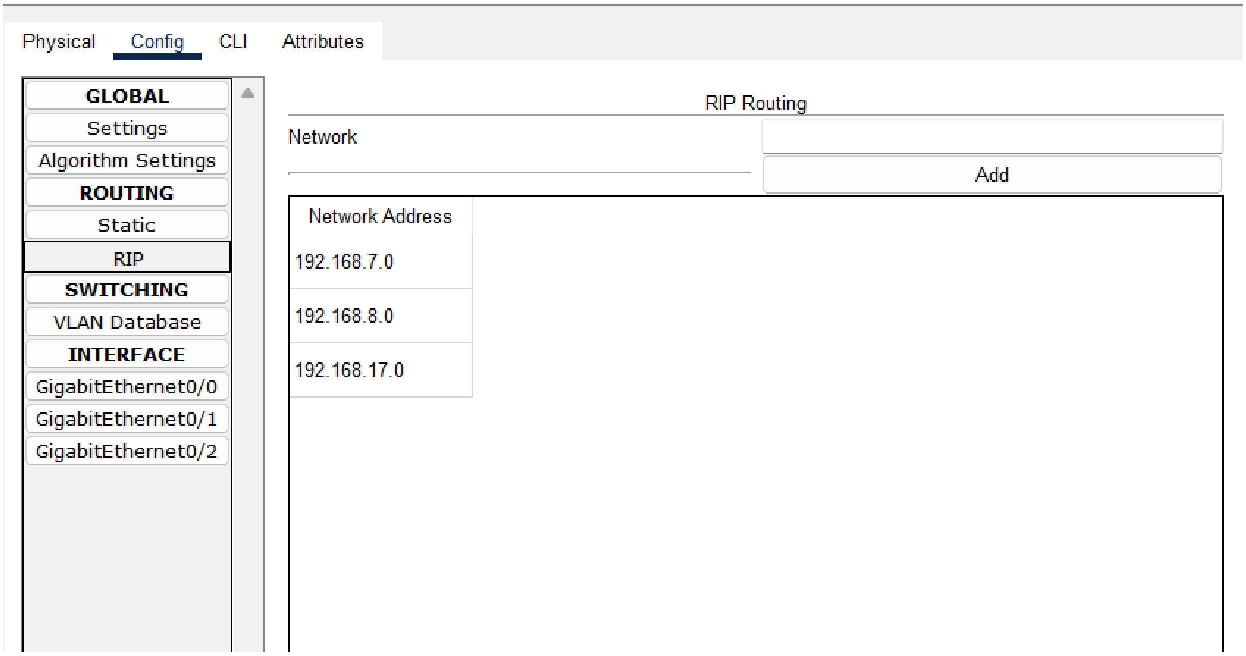
7th router:-



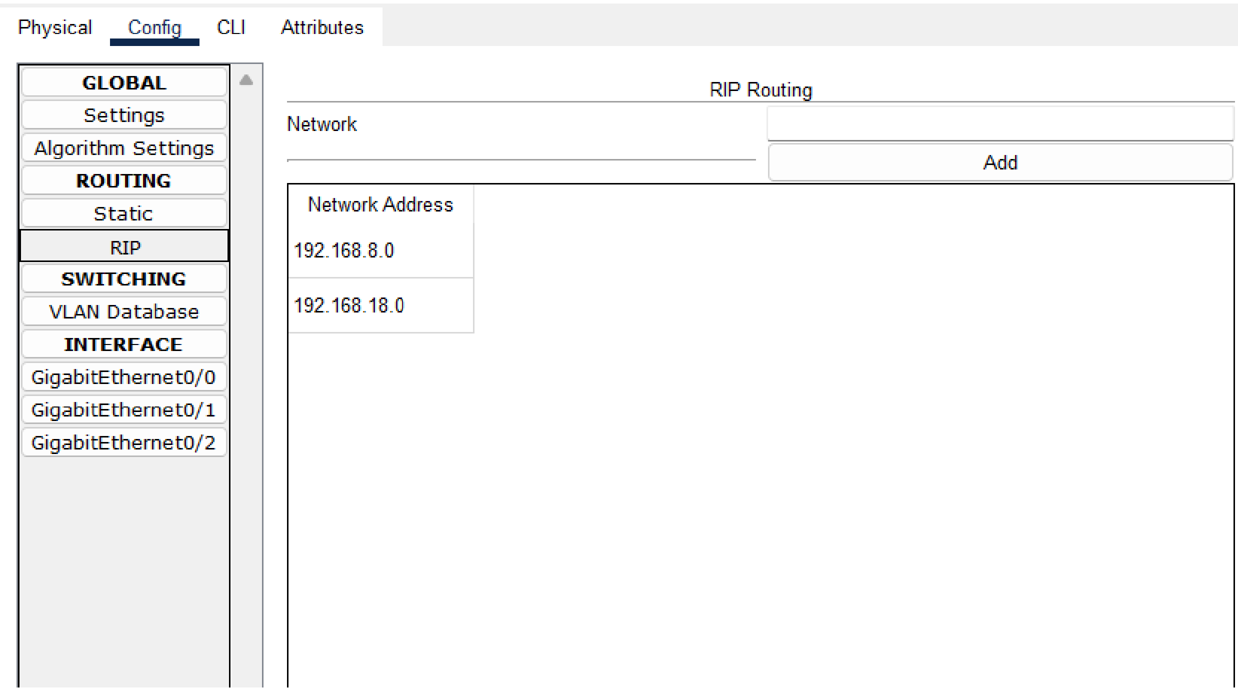
8th router:-



9th router:-



10th router:



Communication between all computers:

1st Floor PC to Floor to all floors:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.5

Pinging 192.168.20.5 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.5: bytes=32 time<1ms TTL=126
Reply from 192.168.20.5: bytes=32 time<1ms TTL=126
Reply from 192.168.20.5: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.20.5:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.11.6

Pinging 192.168.11.6 with 32 bytes of data:

Request timed out.
Reply from 192.168.11.6: bytes=32 time<1ms TTL=125
Reply from 192.168.11.6: bytes=32 time<1ms TTL=125
Reply from 192.168.11.6: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.11.6:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.12.4

Pinging 192.168.12.4 with 32 bytes of data:

Request timed out.
Reply from 192.168.12.4: bytes=32 time<1ms TTL=124
Reply from 192.168.12.4: bytes=32 time<1ms TTL=124
Reply from 192.168.12.4: bytes=32 time=4ms TTL=124

Ping statistics for 192.168.12.4:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 4ms, Average = 1ms

C:\>192.168.13.6
Invalid Command.
```

```
C:\>ping 192.168.13.6

Pinging 192.168.13.6 with 32 bytes of data:

Request timed out.
Reply from 192.168.13.6: bytes=32 time<1ms TTL=123
Reply from 192.168.13.6: bytes=32 time<1ms TTL=123
Reply from 192.168.13.6: bytes=32 time<1ms TTL=123

Ping statistics for 192.168.13.6:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.14.6

Pinging 192.168.14.6 with 32 bytes of data:

Request timed out.
Reply from 192.168.14.6: bytes=32 time=4ms TTL=122
Reply from 192.168.14.6: bytes=32 time<1ms TTL=122
Reply from 192.168.14.6: bytes=32 time=1ms TTL=122

Ping statistics for 192.168.14.6:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 4ms, Average = 1ms

C:\>ping 192.168.15.4

Pinging 192.168.15.4 with 32 bytes of data:

Request timed out.
Reply from 192.168.15.4: bytes=32 time<1ms TTL=121
Reply from 192.168.15.4: bytes=32 time<1ms TTL=121
Reply from 192.168.15.4: bytes=32 time<1ms TTL=121

Ping statistics for 192.168.15.4:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.16.4

Pinging 192.168.16.4 with 32 bytes of data:

Request timed out.
Reply from 192.168.16.4: bytes=32 time<1ms TTL=120
Reply from 192.168.16.4: bytes=32 time<1ms TTL=120
Reply from 192.168.16.4: bytes=32 time<1ms TTL=120
```

9th floor PC to all floor PC's :

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.17.4

Pinging 192.168.17.4 with 32 bytes of data:

Request timed out.
Reply from 192.168.17.4: bytes=32 time<1ms TTL=126
Reply from 192.168.17.4: bytes=32 time=1ms TTL=126
Reply from 192.168.17.4: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.17.4:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.16.3

Pinging 192.168.16.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.16.3: bytes=32 time<1ms TTL=125
Reply from 192.168.16.3: bytes=32 time<1ms TTL=125
Reply from 192.168.16.3: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.16.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.15.4

Pinging 192.168.15.4 with 32 bytes of data:

Reply from 192.168.15.4: bytes=32 time=28ms TTL=124
Reply from 192.168.15.4: bytes=32 time<1ms TTL=124
Reply from 192.168.15.4: bytes=32 time<1ms TTL=124
Reply from 192.168.15.4: bytes=32 time<1ms TTL=124

Ping statistics for 192.168.15.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 28ms, Average = 7ms
```

```
C:\>ping 192.168.13.5

Pinging 192.168.13.5 with 32 bytes of data:

Request timed out.
Reply from 192.168.13.5: bytes=32 time=1ms TTL=122
Reply from 192.168.13.5: bytes=32 time<1ms TTL=122
Reply from 192.168.13.5: bytes=32 time<1ms TTL=122

Ping statistics for 192.168.13.5:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.12.6

Pinging 192.168.12.6 with 32 bytes of data:

Request timed out.
Reply from 192.168.12.6: bytes=32 time<1ms TTL=121
Reply from 192.168.12.6: bytes=32 time<1ms TTL=121
Reply from 192.168.12.6: bytes=32 time<1ms TTL=121

Ping statistics for 192.168.12.6:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.11.4

Pinging 192.168.11.4 with 32 bytes of data:

Request timed out.
Reply from 192.168.11.4: bytes=32 time<1ms TTL=120
Reply from 192.168.11.4: bytes=32 time=1ms TTL=120
Reply from 192.168.11.4: bytes=32 time<1ms TTL=120

Ping statistics for 192.168.11.4:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>192.168.10.3
Invalid Command.
```

```
C:\>ping 192.168.9.2

Pinging 192.168.9.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.9.2: bytes=32 time=24ms TTL=118
Reply from 192.168.9.2: bytes=32 time=10ms TTL=118
Reply from 192.168.9.2: bytes=32 time=1ms TTL=118

Ping statistics for 192.168.9.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 24ms, Average = 11ms

C:\>ping 192.168.20.6

Pinging 192.168.20.6 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.6: bytes=32 time<1ms TTL=119
Reply from 192.168.20.6: bytes=32 time<1ms TTL=119
Reply from 192.168.20.6: bytes=32 time=24ms TTL=119

Ping statistics for 192.168.20.6:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 24ms, Average = 8ms
```